

Method	GSE24080		TCGA LGG methylation	
	Nodes	GPU time (s)	Nodes	GPU time (s)
Exact degree	27,338	1.172 ± 0.035	198,033	66.377 ± 1.196
$\varepsilon = 10^{-1}$	44–53	0.085 ± 0.003	37–49	0.375 ± 0.009
$\varepsilon = 10^{-3}$	46–54	0.089 ± 0.010	39–51	0.374 ± 0.019
$\varepsilon = 10^{-6}$	48–57	0.086 ± 0.008	41–53	0.385 ± 0.018

Table 1: Cox explanation times on an Apple M4 Pro GPU (JAX-Metal, FP32), using 30 fixed training-background samples. GSE24080 and TCGA LGG have $(n_{\text{train}}, d) = (339, 54,675)$ and $(383, 396,065)$, respectively. Times are mean $\pm$ sample standard deviation across five distinct held-out patients, with one synchronized call per patient and method. Node ranges span patients. Warm-up and cached rule construction are excluded; automatic node selection is included. “Exact degree” denotes exact quadrature in real arithmetic; the requested absolute quadrature tolerance ε excludes FP32 rounding. Measured absolute errors exceeded the requested tolerance in 19/30 approximate cases. Previously measured CPU construction of the cached exact rules took 10.97 s (GSE24080) and 561.92 s (TCGA). Fresh approximate-rule construction took 0.418–0.625 ms (range of per-rule medians over five constructions).